



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Javier Rodriguez
May 1, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

All processes addressed in this project are supported by the CRISP-DM or Cross-Industry Standard Process for Data Mining methodology. They are based on the following steps:

- First, we will understand the Business: We will define what problem we want to solve and what objectives we are seeking.
- Next, we will collect the Data: We will gather all the data from various sources.
- Then, we will prepare the Data: Cleaning, transformation, and variable selection.
- Next, we will understand the Data: We will explore the data quality through exploratory data analysis (EDA) with SQL and visualization.
- Next, we will model the Data: We will apply Machine Learning algorithms.
- Finally, we will evaluate the models: We will review whether the model meets the business objectives.

Introduction

We will predict if the Falcon 9 first stage will land successfully. SpaceX advertises Falcon 9 rocket launches on its website, with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage. Therefore, if we can determine whether the first stage will land, we can determine the cost of a launch. This information can be used if an alternative company wants to bid against SpaceX for a rocket launch.

From this point, we will discover:

- Which are the main variables that control these successful landings, maybe the flight number, the payload mass or others?
- What is the location of all launch sites and why were they chosen?
- Which specific site has the highest success rate and what is the ideal payload range?
- Which Machine Learning model performs best and what are their limitations?

Section 1

Methodology

Methodology - Executive Summary – Part 1

- Data collection methodology:
 - Data was collected using SpaceX REST API and Web Scraping from Wikipedia.
- Perform data wrangling
 - The data was preprocessed using Pandas and NumPy, for example One-hot encoding was applied to categorical features.
- Perform exploratory data analysis (EDA) using visualization and SQL
 - Using libraries such as seaborn and matplotlib for data visualization and SQL for data queries.

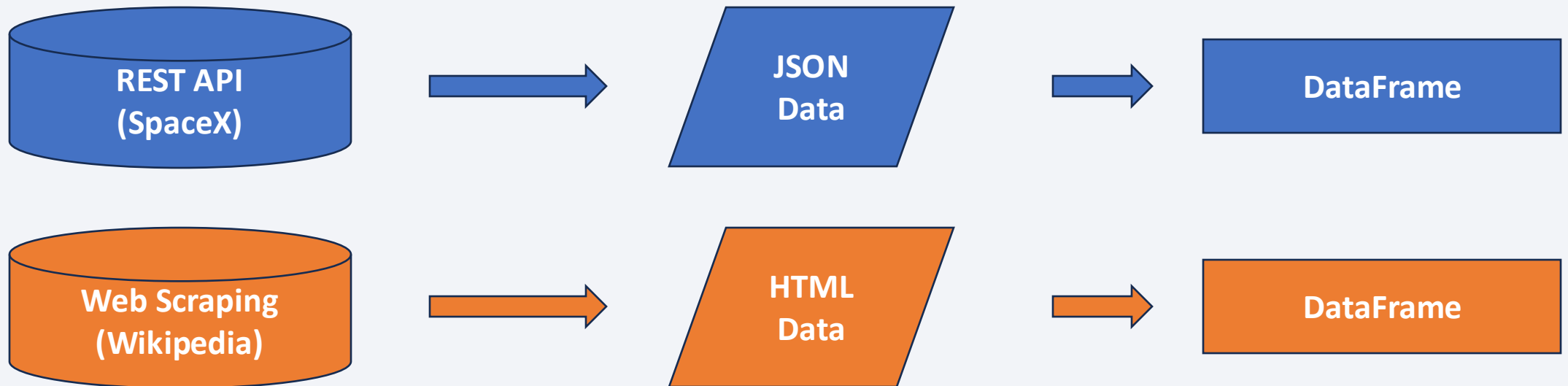
Methodology - Executive Summary – Part 2

- Perform interactive visual analytics using Folium and Plotly Dash
 - Using libraries such as Folium for Interactive Visual Analytics and Plotly Dash to build an Interactive app.
- Perform predictive analysis using classification models
 - Starting with splitting the data into train and test sets, then using Grid Search to find best algorithm and parameters and evaluating the classification models with a Confusion Matrix.

Data Collection

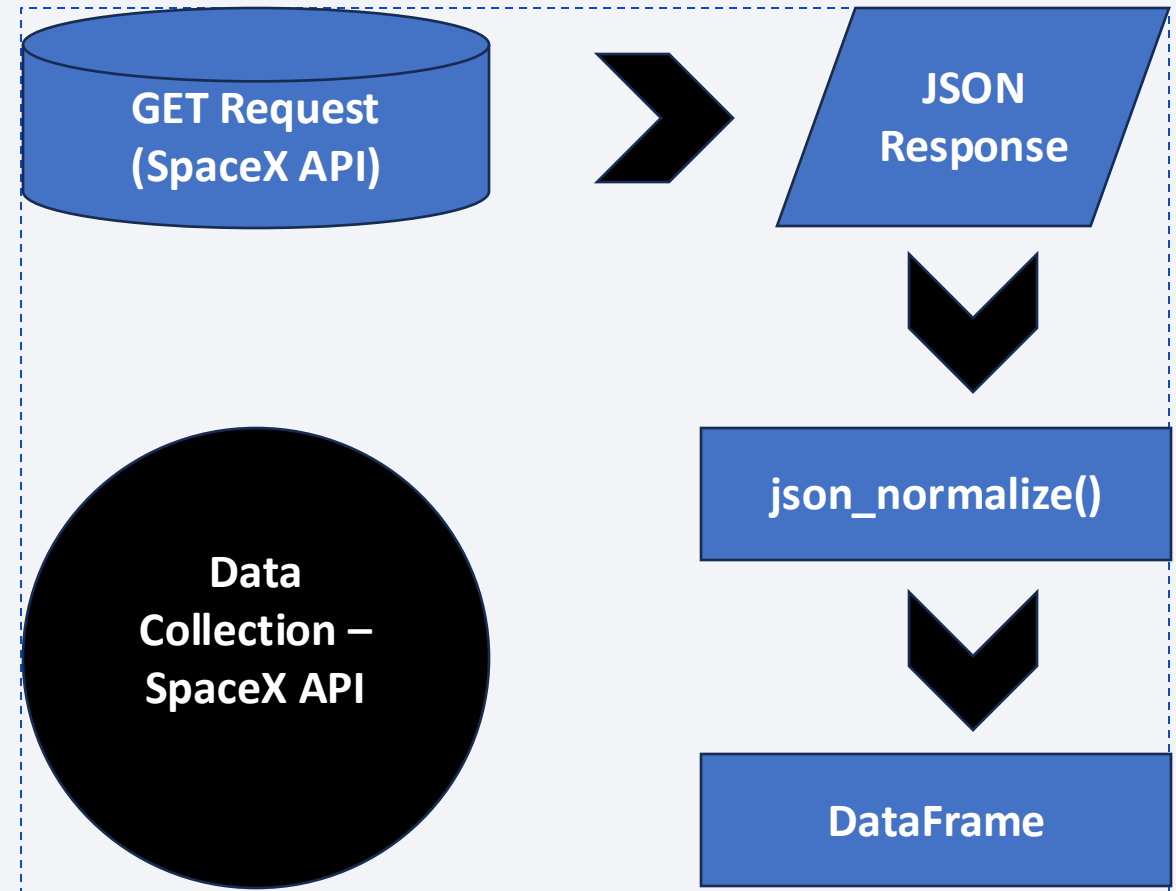
- The data was collected from two sources: SpaceX REST API and Wikipedia using Web Scraping.

Here is the data collection process:



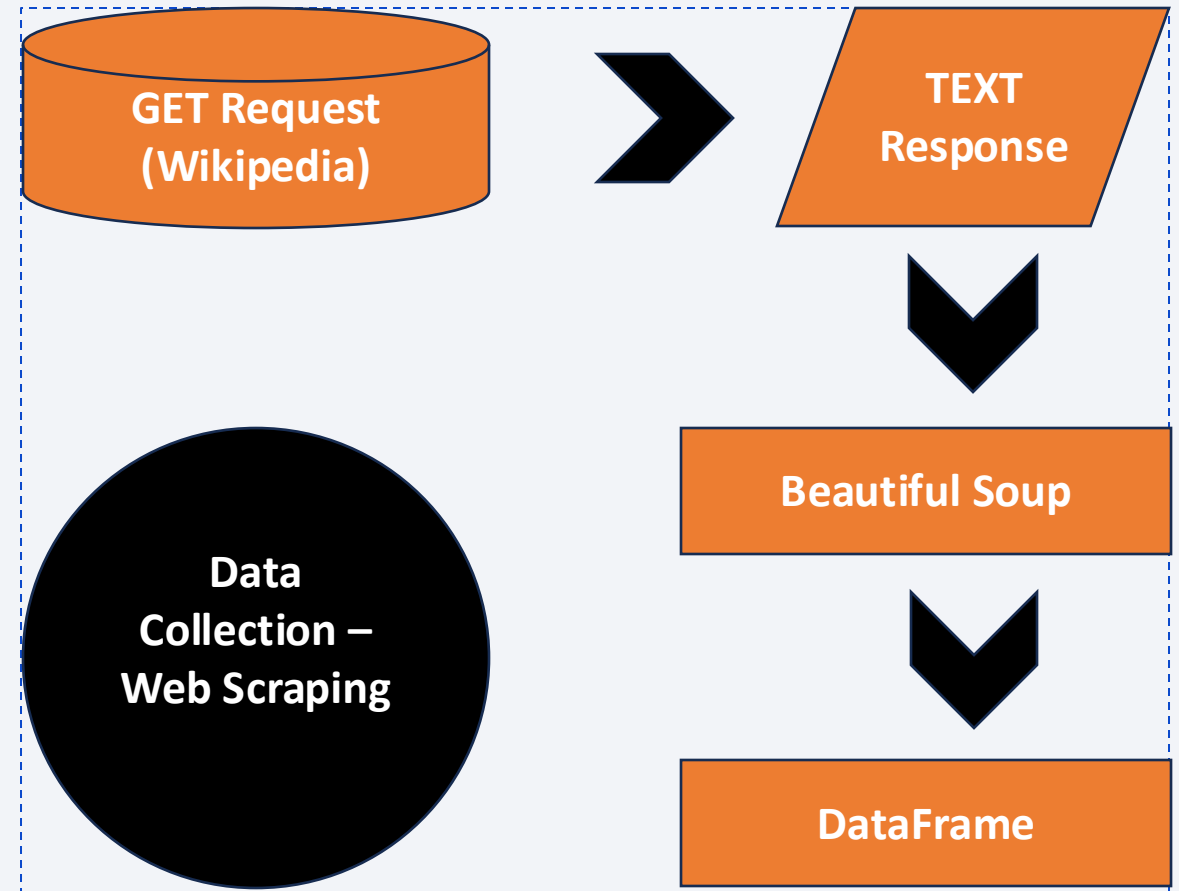
Data Collection – SpaceX API

- We started the data collection by getting a URL request to the SpaceX API, this request is raised as JSON file to be finally converted to a data frame using `json_normalize()` method.
- You can check my GitHub URL of the completed SpaceX API calls notebook here: [Data Collection API](#)



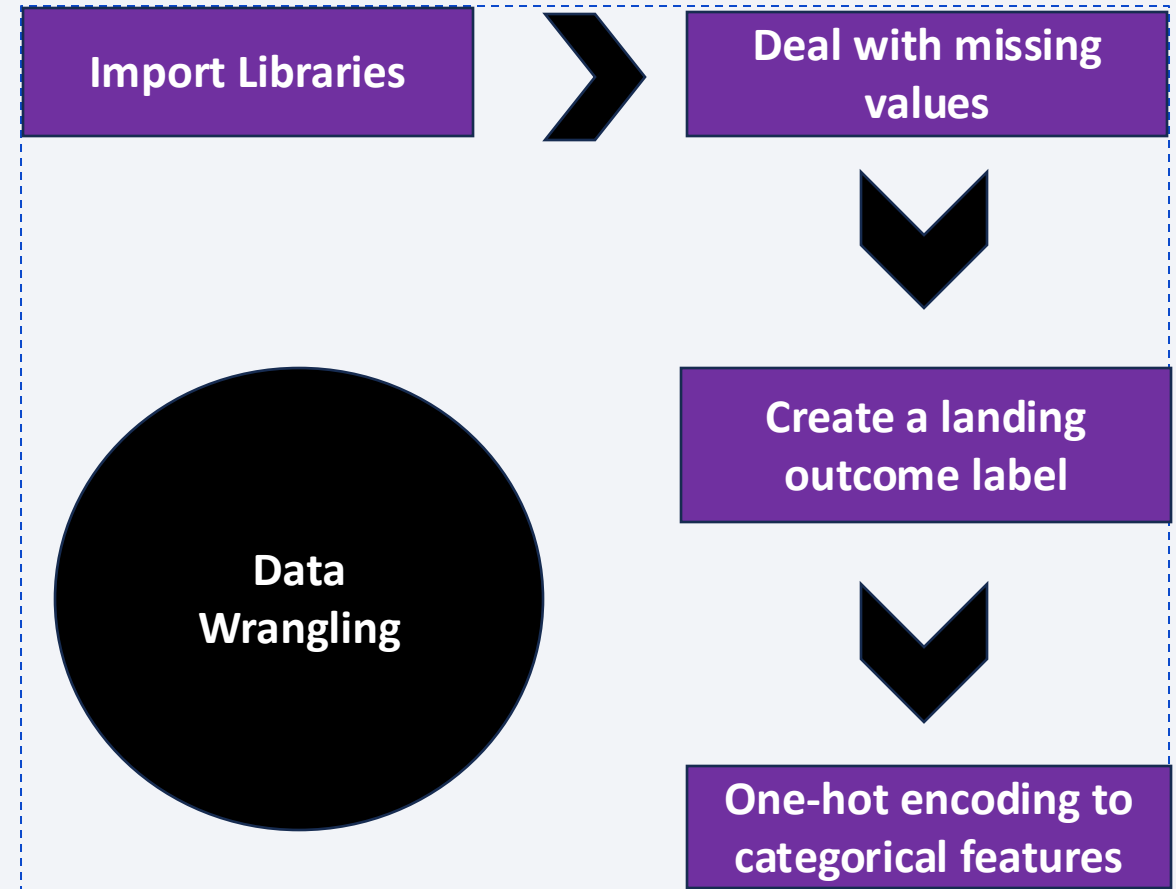
Data Collection - Scraping

- We continued our process of data collection with Web Scraping by performing an HTTP Get Request to the Falcon9 Launch HTML page, this request is raised as response text format and with a BeautifulSoup object, we extract all column/variable names from the text response to be converted later to a data frame.
- You can check my GitHub URL of the completed web scraping notebook here: [Data Collection with Web Scraping](#)



Data Wrangling

- The data was preprocessed using Pandas and NumPy. We first deal with missing values, then we create a landing outcome label from Outcome column and finally One-hot encoding was applied to categorical features.
- You can check my GitHub URL of the completed data wrangling notebook here: [Data Wrangling](#)



Dealing with missing values

```
data_falcon9.isnull().sum()
```

```
FlightNumber      0
Date              0
BoosterVersion    0
PayloadMass       5
Orbit             0
LaunchSite        0
Outcome           0
Flights           0
GridFins          0
Reused            0
Legs              0
LandingPad       26
Block             0
ReusedCount       0
Serial            0
Longitude         0
Latitude          0
dtype: int64
```

```
mean = data_falcon9['PayloadMass'].mean()
```

```
data_falcon9['PayloadMass'].replace(np.nan, mean,
                                     inplace=True)
```

- After seeing that columns PayloadMass and LandingPad have missing values, we will calculate the mean for the PayloadMass column and replace it with the missing values.
- The LandingPad column will retain None values to represent when landing pads were not used.

Creating a landing outcome label Part 1

- This time, we convert original outcome labels into landing class that represent landing classification which will be our new landing prediction target.

0 True ASDS = 1

1 None None = 0

2 True RTLS = 1

3 False ASDS = 0

4 True Ocean = 1

5 False Ocean = 0

6 None ASDS = 0

7 False RTLS = 0

Creating a landing outcome label Part 2

- Now, we create a list of the elements, where the element is zero if the corresponding row in Outcome is in the set bad_outcome; otherwise, it's one. Then assign it to the variable landing_class and finally assign the variable in a new column named Class.

```
bad_outcomes=set(landing_outcomes.keys()[[1,3,5,6,7]])  
bad_outcomes
```

```
{'False ASDS', 'False Ocean', 'False RTLS', 'None ASDS', 'None None'}
```

```
landing_class = []  
# landing_class = 0 if bad_outcome  
for row in df['Outcome']:  
    if row in bad_outcomes:  
        landing_class.append(0)  
# landing_class = 1 otherwise  
else:  
    landing_class.append(1)
```

```
df['Class']=landing_class  
df[['Class']].head(8)
```

Class	
0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	1

One-hot encoding to categorical features Part 1

- From a select features that will be used in success prediction, we apply One-hot encoding to the columns Orbits, LaunchSite, Landing Pad and Serial.
- Then we assign them in the variable features_one_hot.

	FlightNumber	PayloadMass	Orbit	LaunchSite	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial
0	1	6104.959412	LEO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0003
1	2	525.000000	LEO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0005
2	3	677.000000	ISS	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0007
3	4	500.000000	PO	VAFB SLC 4E	1	False	False	False	NaN	1.0	0	B1003
4	5	3170.000000	GTO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B1004

One-hot encoding to categorical features Part 2

```
dummies = pd.get_dummies(features[['Orbit', 'LaunchSite', 'LandingPad', 'Serial']]).astype(int)
```

```
features_one_hot = pd.concat([features, dummies], axis=1)
```

```
features_one_hot = features_one_hot.drop(['Orbit', 'LaunchSite', 'LandingPad', 'Serial'], axis=1)  
features_one_hot.head()
```

	FlightNumber	PayloadMass	Flights	GridFins	Reused	Legs	Block	ReusedCount	Orbit_ES- L1	Orbit_GEO	...	Serial_B1048	Serial_B1049
0	1	6104.959412	1	False	False	False	1.0	0	0	0	...	0	0
1	2	525.000000	1	False	False	False	1.0	0	0	0	...	0	0
2	3	677.000000	1	False	False	False	1.0	0	0	0	...	0	0
3	4	500.000000	1	False	False	False	1.0	0	0	0	...	0	0
4	5	3170.000000	1	False	False	False	1.0	0	0	0	...	0	0

5 rows × 80 columns

One-hot encoding to categorical features Part 3

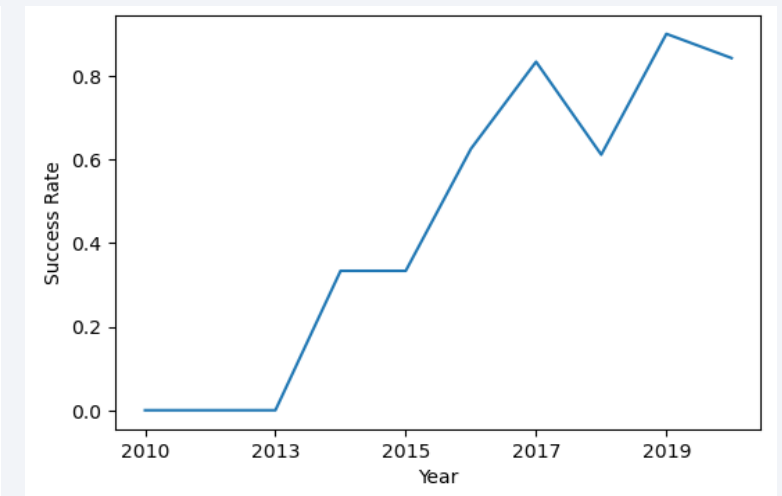
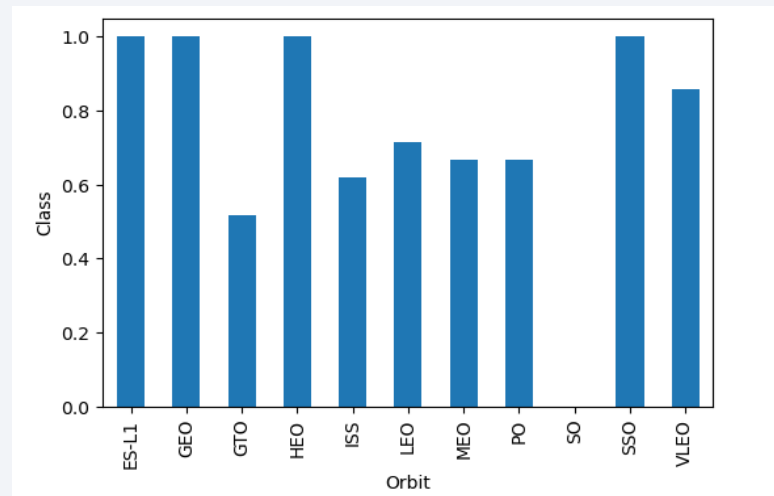
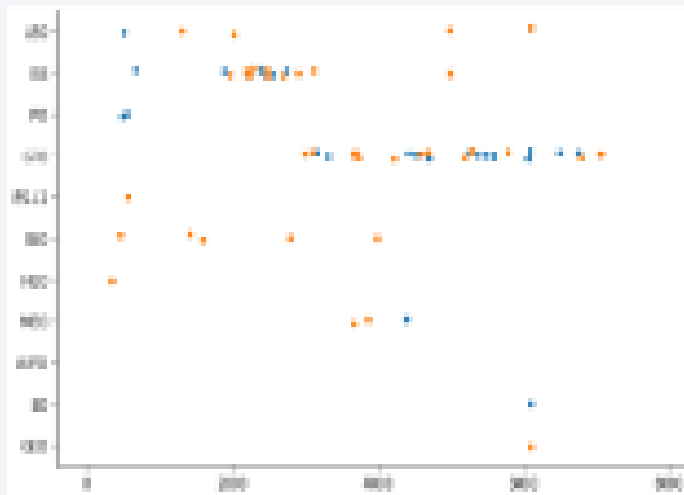
- Finally we cast the entire DataFrame to variable type float64.

	FlightNumber	PayloadMass	Flights	GridFins	Reused	Legs	Block	ReusedCount	Orbit_ES-L1	Orbit_GEO	...	Serial_B1048	Serial_B1049
0	1.0	6104.959412	1.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	...	0.0	0.0
1	2.0	525.000000	1.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	...	0.0	0.0
2	3.0	677.000000	1.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	...	0.0	0.0
3	4.0	500.000000	1.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	...	0.0	0.0
4	5.0	3170.000000	1.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	...	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
85	86.0	15400.000000	2.0	1.0	1.0	1.0	5.0	2.0	0.0	0.0	...	0.0	0.0
86	87.0	15400.000000	3.0	1.0	1.0	1.0	5.0	2.0	0.0	0.0	...	0.0	0.0
87	88.0	15400.000000	6.0	1.0	1.0	1.0	5.0	5.0	0.0	0.0	...	0.0	0.0
88	89.0	15400.000000	3.0	1.0	1.0	1.0	5.0	2.0	0.0	0.0	...	0.0	0.0
89	90.0	3681.000000	1.0	1.0	0.0	1.0	5.0	0.0	0.0	0.0	...	0.0	0.0

90 rows × 80 columns

EDA with Data Visualization

- In the exploratory data analysis with visualization, we use scatter plots to get relationship between variables, a bar plot to plot success rate of each orbit and a line chart to get the yearly average launch success trend.
- You can check my GitHub URL of the completed EDA with data visualization notebook here: [EDA with Visualization](#)



EDA with SQL

- ✓ Display the names of the unique launch sites in the space mission.
- ✓ List the date when the first successful landing outcome in ground pad was achieved.
- ✓ List the total number of successful and failure mission outcomes.
- ✓ List the records which will display the month names, failure landing_outcomes in drone ship, booster versions, launch_site for the months in year 2015.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

First_Date
2015-12-22

Success
100

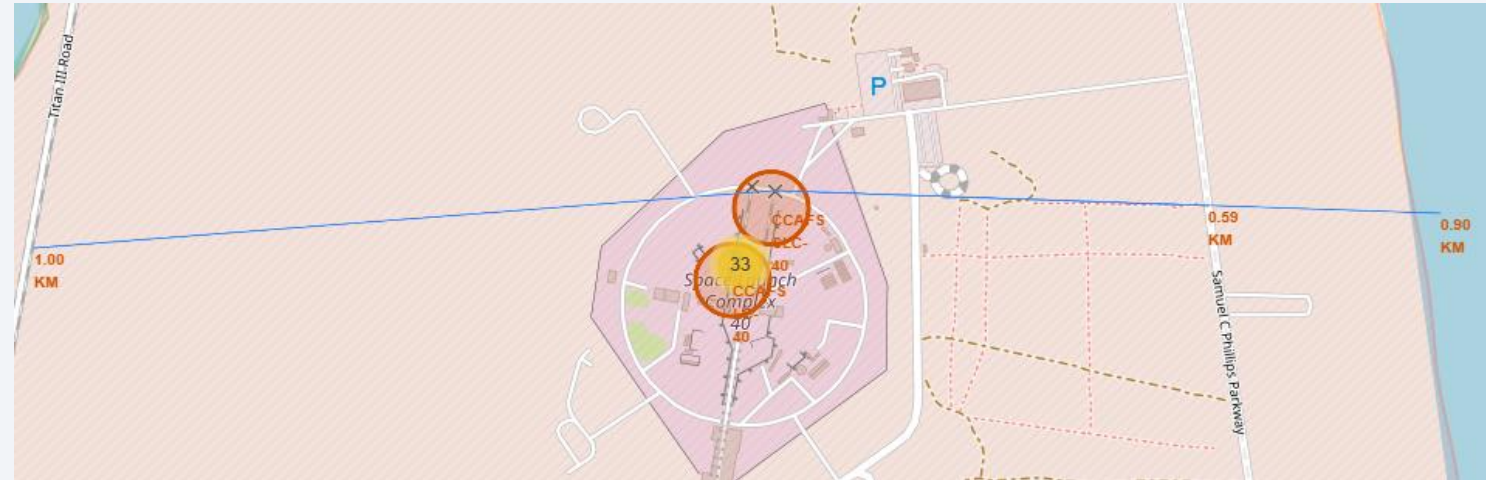
Failure
1

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

You can check my GitHub URL of the completed EDA with SQL notebook here: [EDA with SQL](#)

Build an Interactive Map with Folium

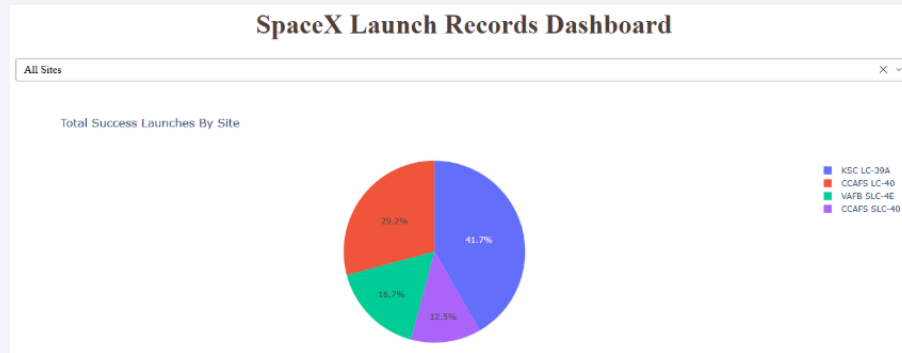
- With this folium map, we create map objects such as markers for labels, circles for launch sites, lines for calculate distances between a launch site and its proximities and marker clusters for successful and failed launches.



- You can check my GitHub URL of the completed interactive map with Folium map notebook here: [Interactive Visual Analytics with Folium](#)

Build a Dashboard with Plotly Dash

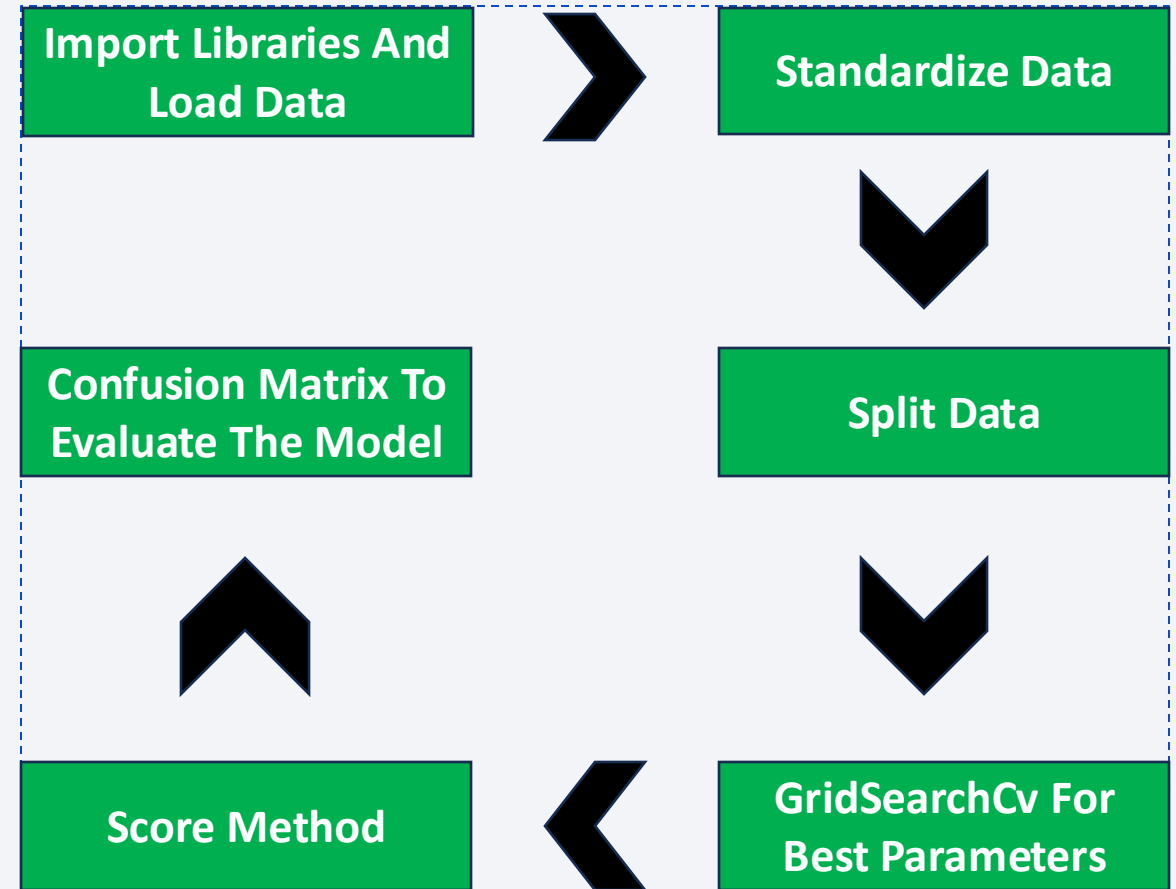
- We added a dropdown menu and a Pie chart to show the total successful launches for all sites and for each launch site.
- Additionally, a Range Slider to select payload which ranges from 0-10000 and a Scatter Plot to show the correlation between Payload and Launch Success for all sites and for each launch site.

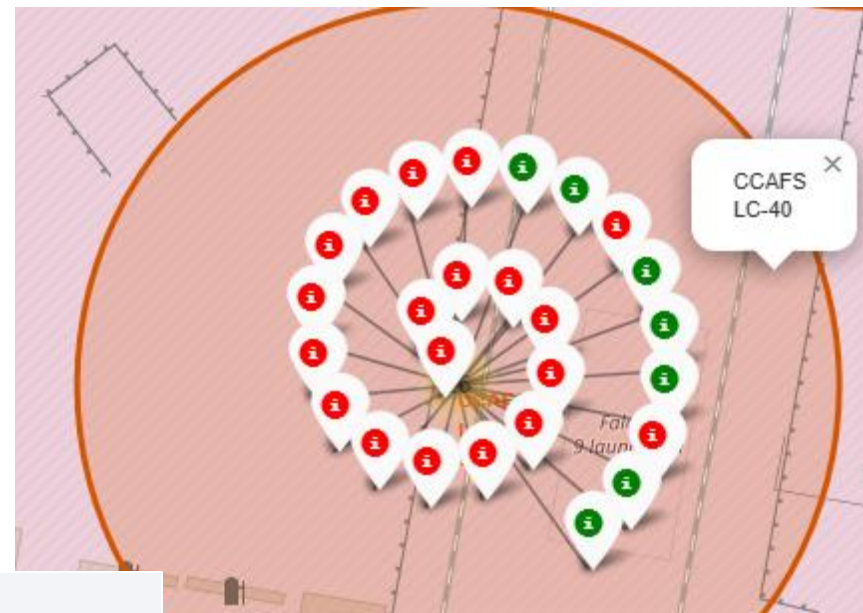
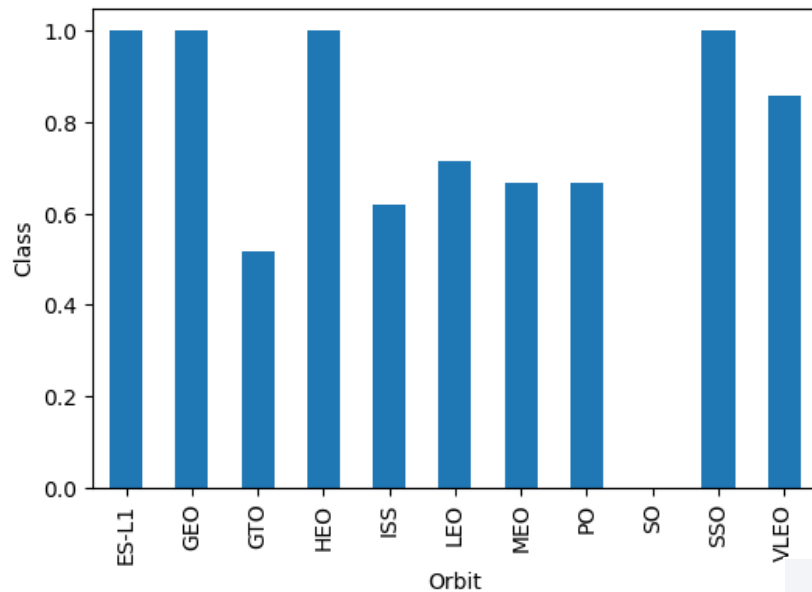


- You can check my GitHub URL of the completed Plotly Dash notebook here: [Spacex Dash App](#)

Predictive Analysis (Classification)

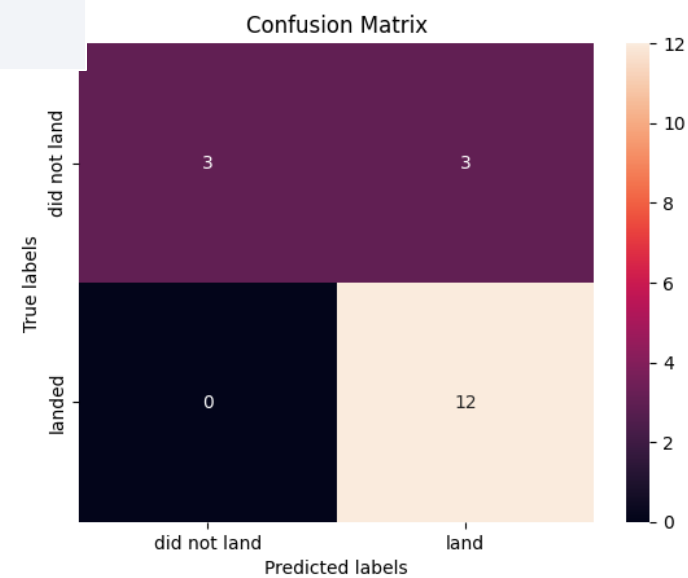
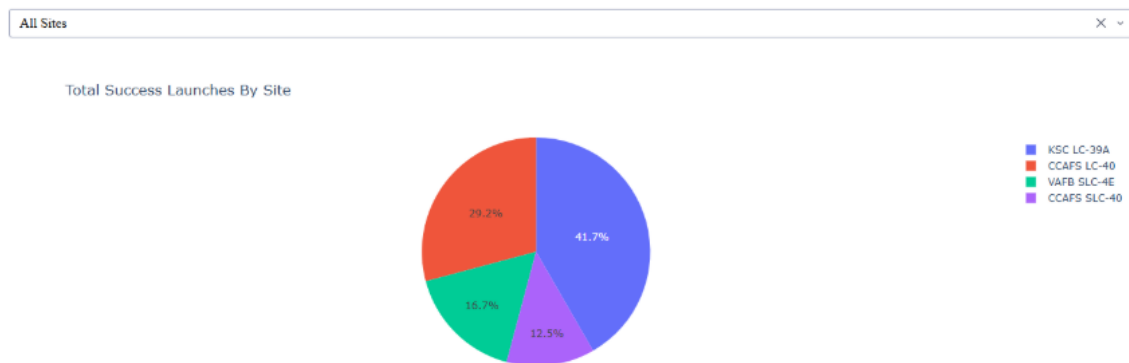
- First, we import the required libraries, then load the data. After that, standardize the data to prevent the bias.
- Next, we split the data 80% for training and 20% for testing data.
- In all models, we used the function GridSearchCV to select best hyperparameters and the method score to calculate the accuracy on the test data.
- Finally, we evaluated the models with evaluation techniques like Confusion Matrix.
- You can check my GitHub URL of the completed predictive analysis notebook here: [Machine Learning Prediction](#)

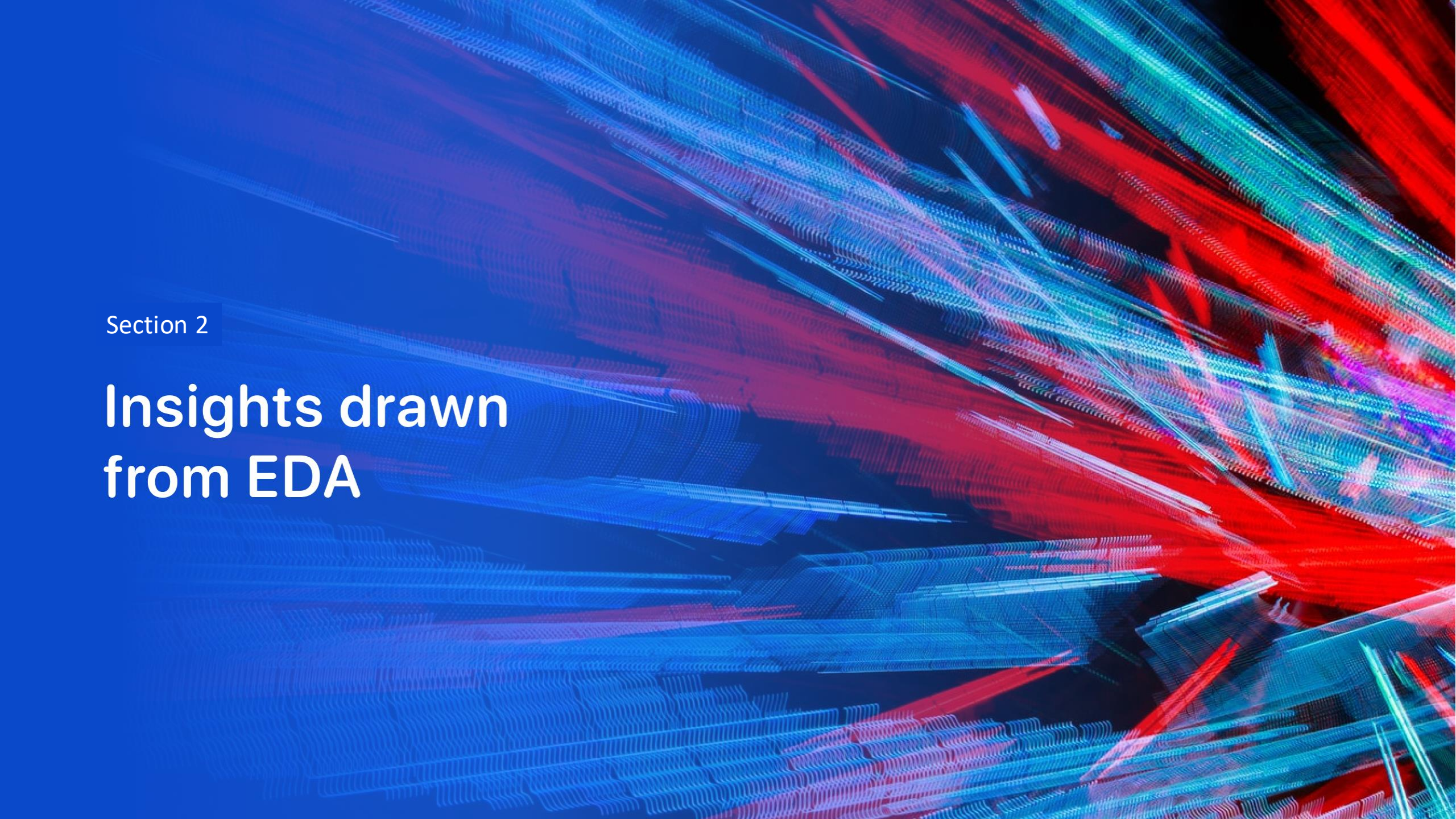




Results

SpaceX Launch Records Dashboard

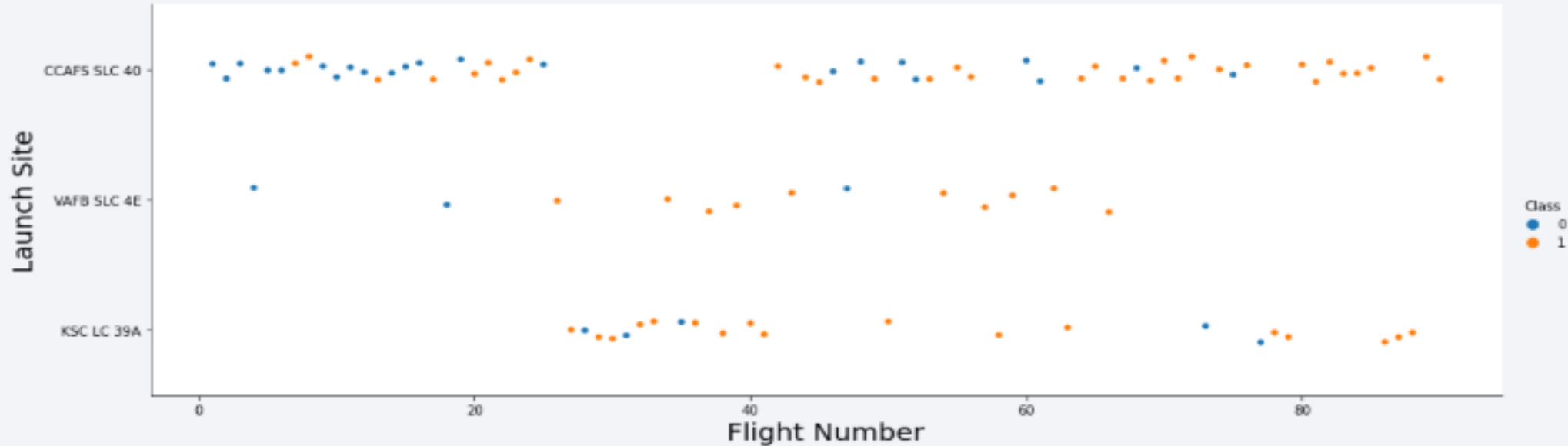


The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

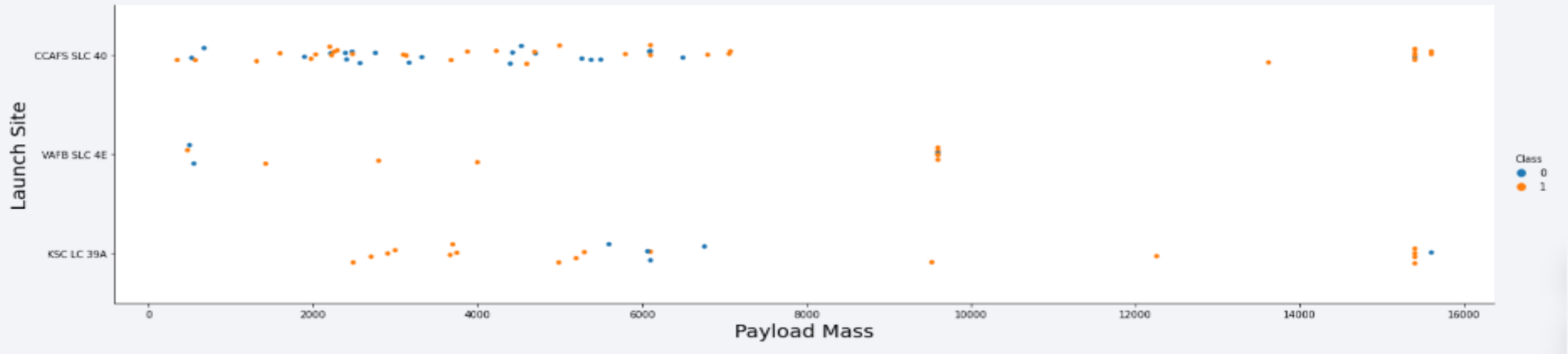
Insights drawn from EDA

Flight Number vs. Launch Site



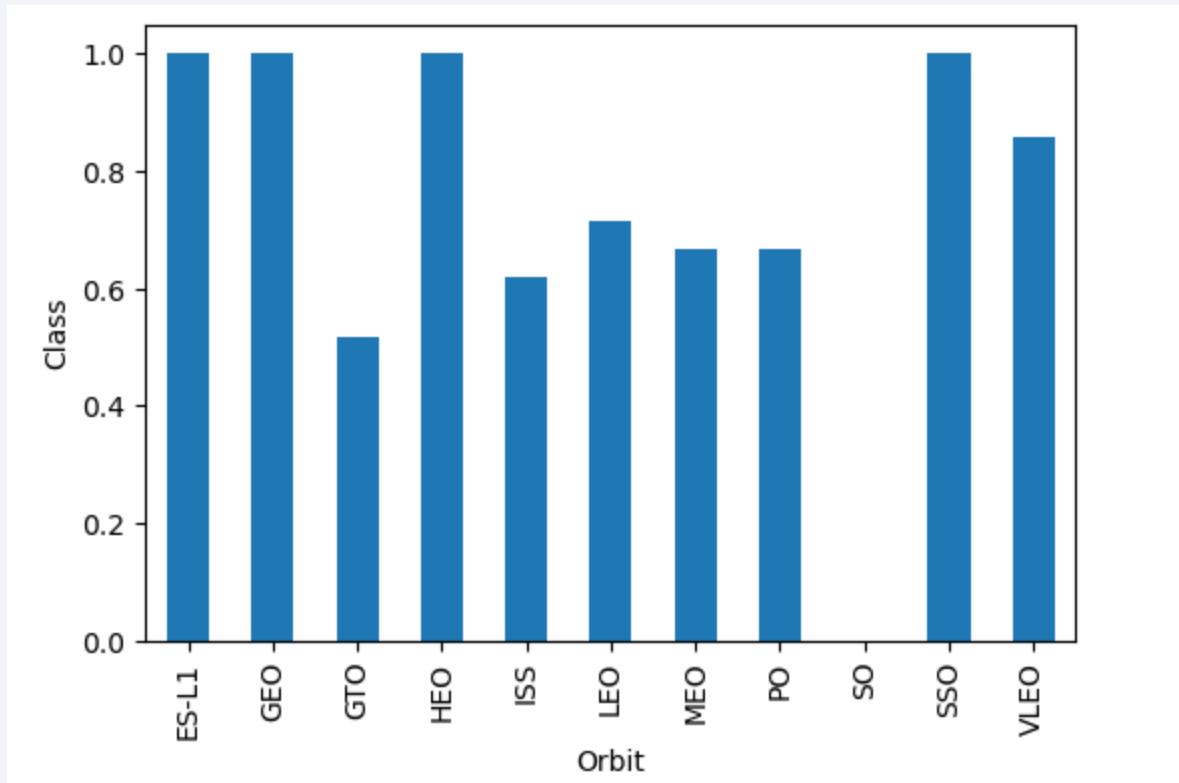
- According to the plot above, you can observe that all launch sites with a higher volume of flights correlates with a success rate.

Payload vs. Launch Site



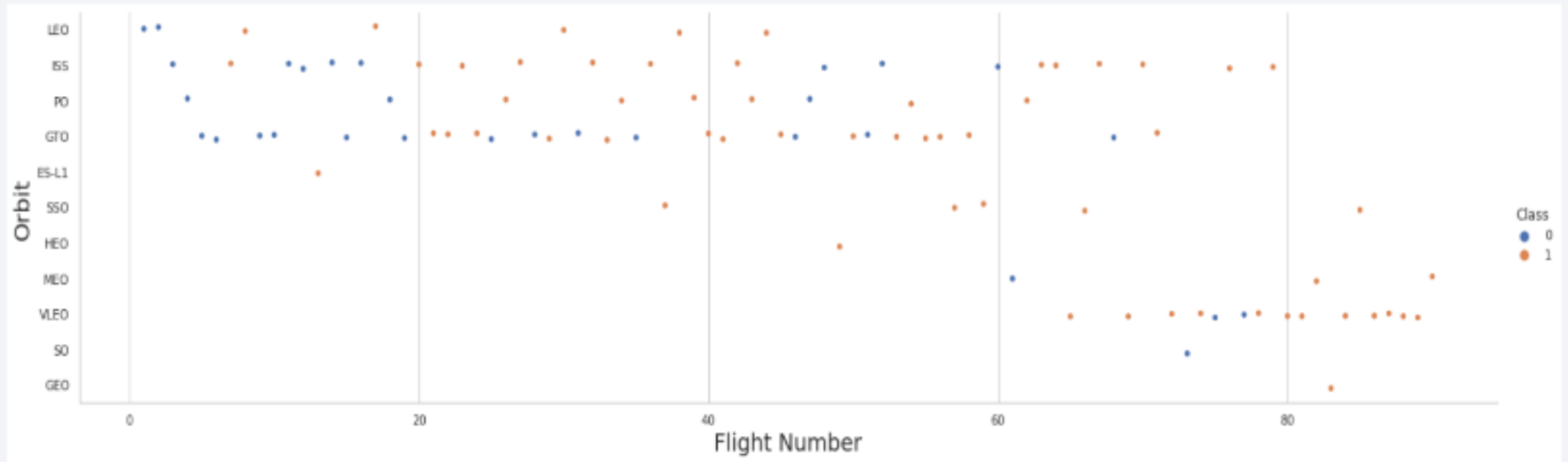
- Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC 4E launch site there are no rockets launched for heavy payload mass (greater than 10000).

Success Rate vs. Orbit Type



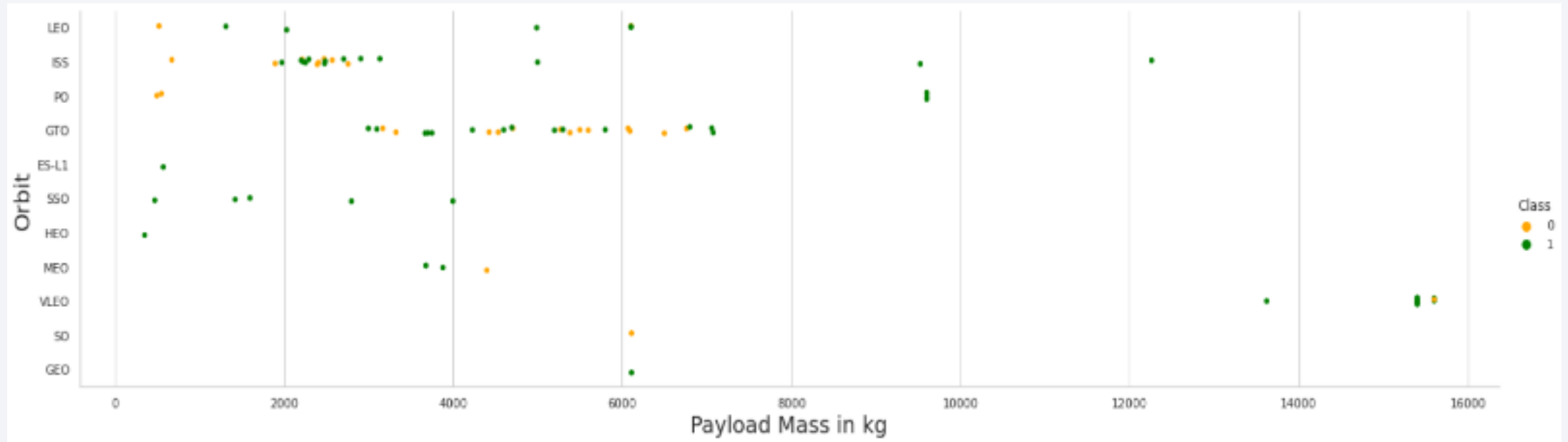
- From the bar chart, we can see that ES-L1, GEO, HEO and SSO orbits have the highest success rates.

Flight Number vs. Orbit Type



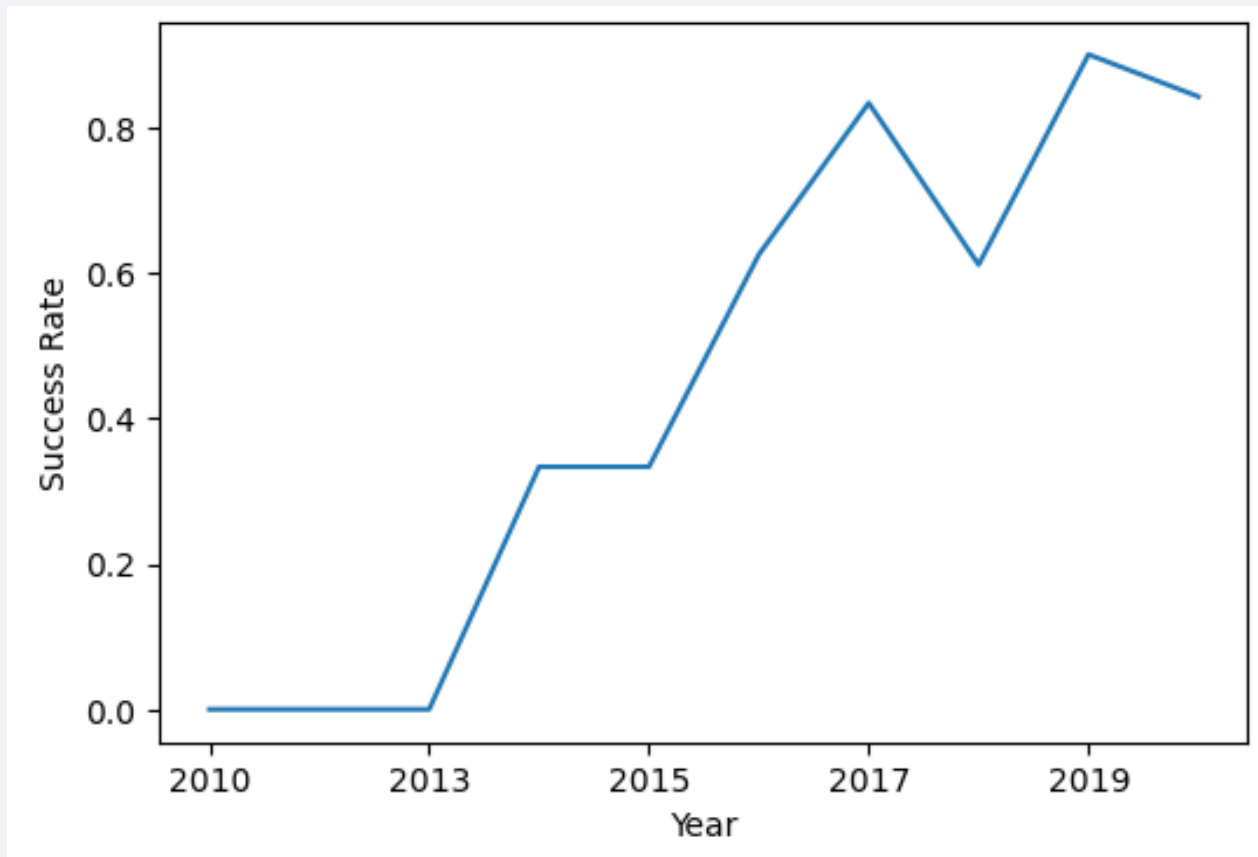
- You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

Payload vs. Orbit Type



- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend



- You can observe that the success rate since 2013 kept increasing till 2020.

All Launch Site Names

```
%sql select DISTINCT Launch_Site from SPACEXTABLE
* sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- According to the results, there are four distinct launch sites.

Launch Site Names Begin with 'CCA'

```
%sql select * from SPACEXTABLE where Launch_Site LIKE 'CCA%' limit 5;
```

```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Here are 5 records where the launch sites begin with the string "CCA"

Total Payload Mass

```
%sql select SUM(PAYLOAD_MASS__KG_) as Total_Payload_Mass from SPACEXTABLE where Customer = 'NASA (CRS)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Total_Payload_Mass

45596

- The total payload mass carried by boosters launched by NASA (CRS) is 45596 kg.

Average Payload Mass by F9 v1.1

```
%sql select AVG(PAYLOAD_MASS__KG_) as Average_Payload_Mass from SPACEXTABLE where Booster_Version = 'F9 v1.1'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Average_Payload_Mass

2928.4

- The average payload mass carried by booster version F9 v1.1 is 2928.4 kg.

First Successful Ground Landing Date

```
%sql select MIN(Date) as First_Date from SPACEXTABLE where Landing_Outcome = 'Success (ground pad)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

First_Date
2015-12-22

- The date of the first successful landing outcome on ground pad was in 2015-12-22.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql select Booster_Version from SPACEXTABLE \
where Landing_Outcome = 'Success (drone ship)' and (PAYLOAD_MASS_KG_ between 4001 and 5999);

* sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

The boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 are:

- F9 FT B1022
- F9 FT B1026
- F9 FT B1021.2
- F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

```
%sql select COUNT(*) as Success from SPACEXTABLE where Mission_Outcome LIKE 'Success%'
* sqlite:///my_data1.db
Done.
```

Success
100

```
%sql select COUNT(*) as Failure from SPACEXTABLE where Mission_Outcome LIKE 'Failure%'
* sqlite:///my_data1.db
Done.
```

Failure
1

- The total number of successful mission outcomes is 100 and the total number of failure mission outcomes is 1.

Boosters Carried Maximum Payload

```
%sql select Booster_Version from SPACEXTABLE \
where PAYLOAD_MASS_KG_ in (select MAX(PAYLOAD_MASS_KG_) from SPACEXTABLE)
```

```
* sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- The booster versions which have carried the maximum payload mass starts with F9 B5 and ranges from B1048 up to B1060.

2015 Launch Records

```
%sql select substr(Date, 6,2) as Month, Landing_Outcome, Booster_Version, Launch_Site from SPACEXTABLE \
where Landing_Outcome = 'Failure (drone ship)' and substr(Date,0,5) = '2015'
```

```
* sqlite:///my_data1.db
Done.
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- There are two failed landings on a drone ship, both from the same launch site and booster version F9 v1.1 in January and April of 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql select Landing_Outcome, COUNT(*) as Total from SPACEXTABLE \
where (Date between '2010-06-04' and '2017-03-20') \
group by Landing_Outcome \
order by Total desc
```

* sqlite:///my_data1.db

Done.

Landing_Outcome	Total
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

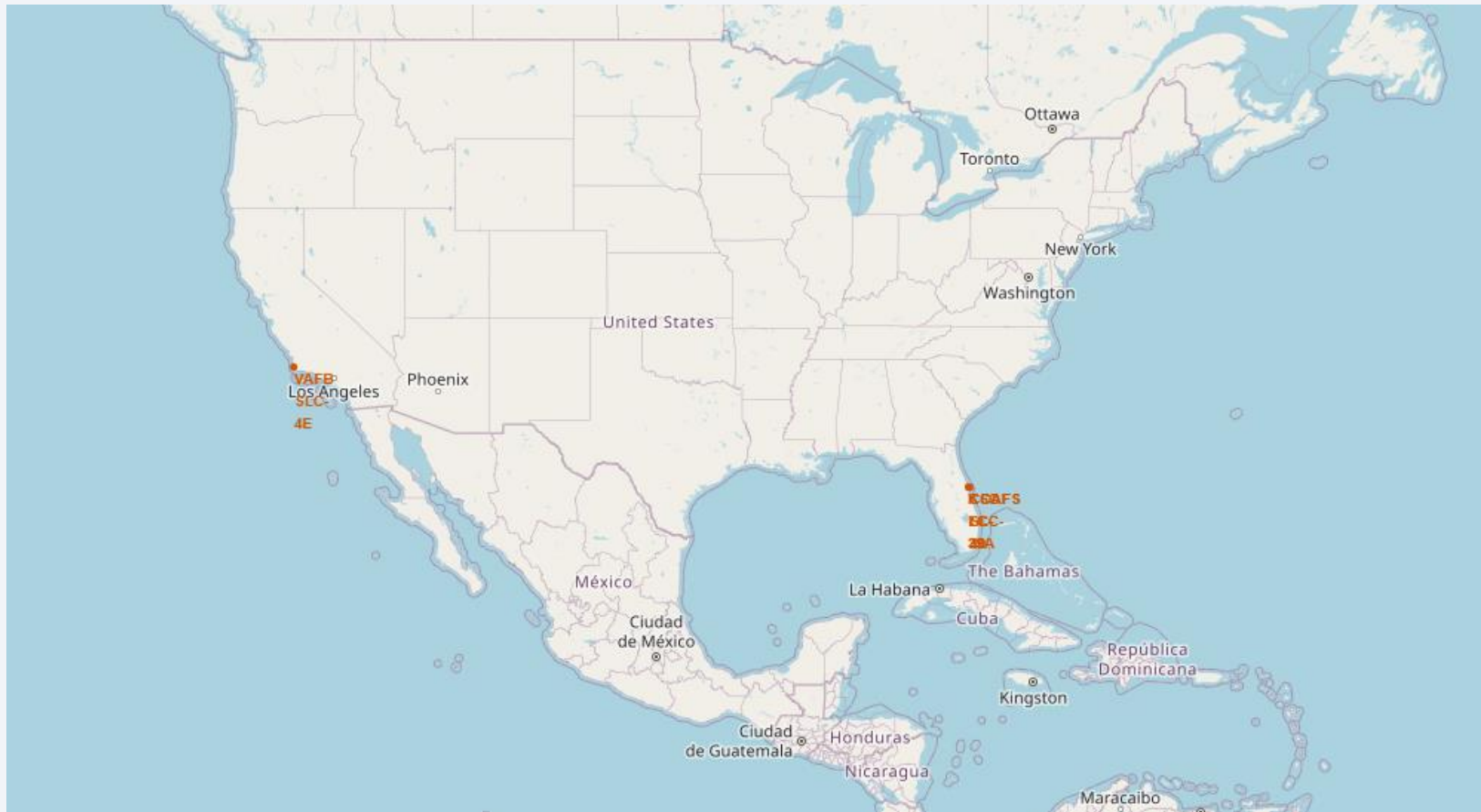
- This is the result of total landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the deep blue of space.

Section 3

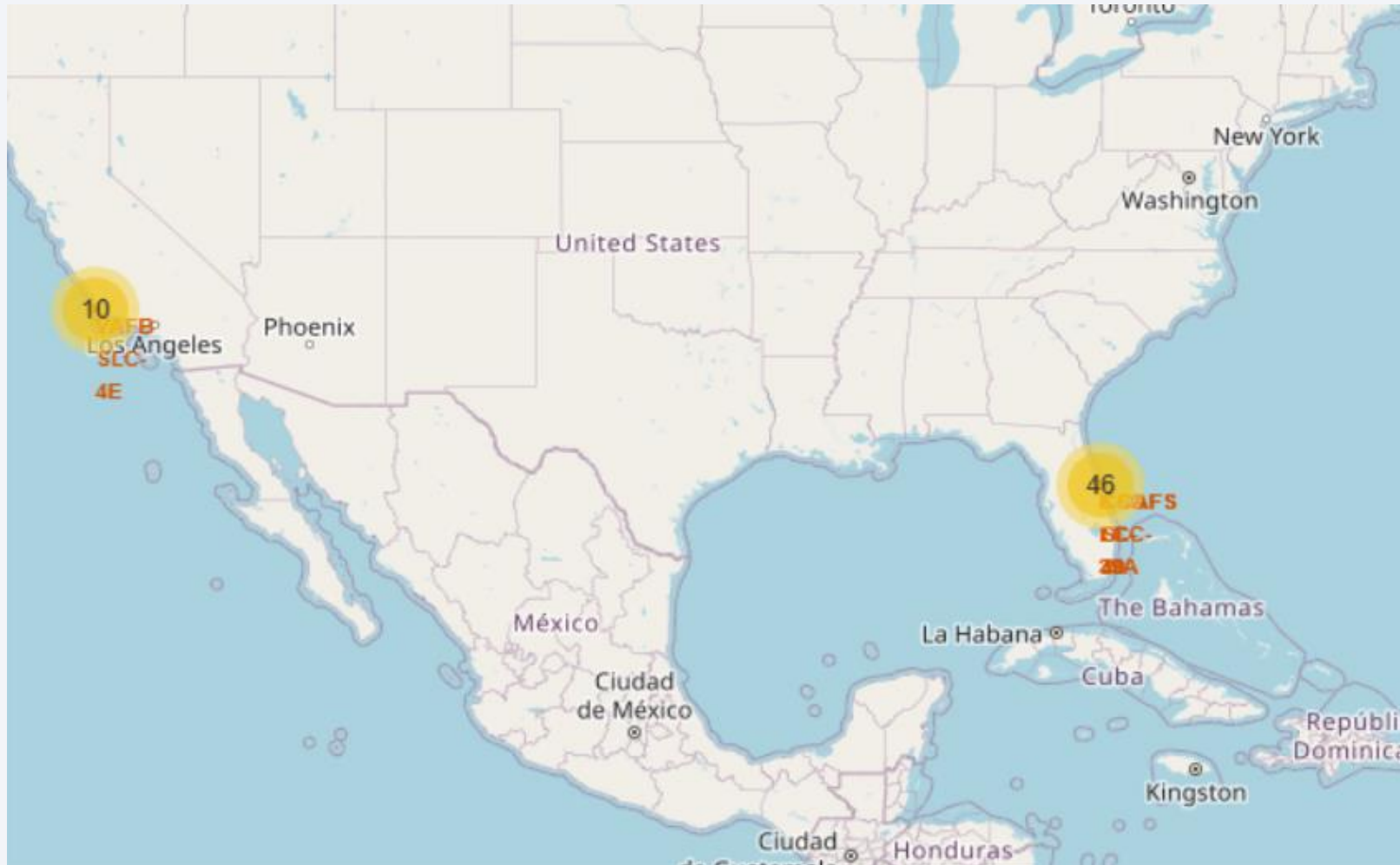
Launch Sites Proximities Analysis

Folium Map: Locations of All Launch Sites



- The launch sites are located in two US states: California and Florida.
- All launch sites are very close to the coast because it allows for safer rocket debris disposal

Folium Map: Marker Clusters



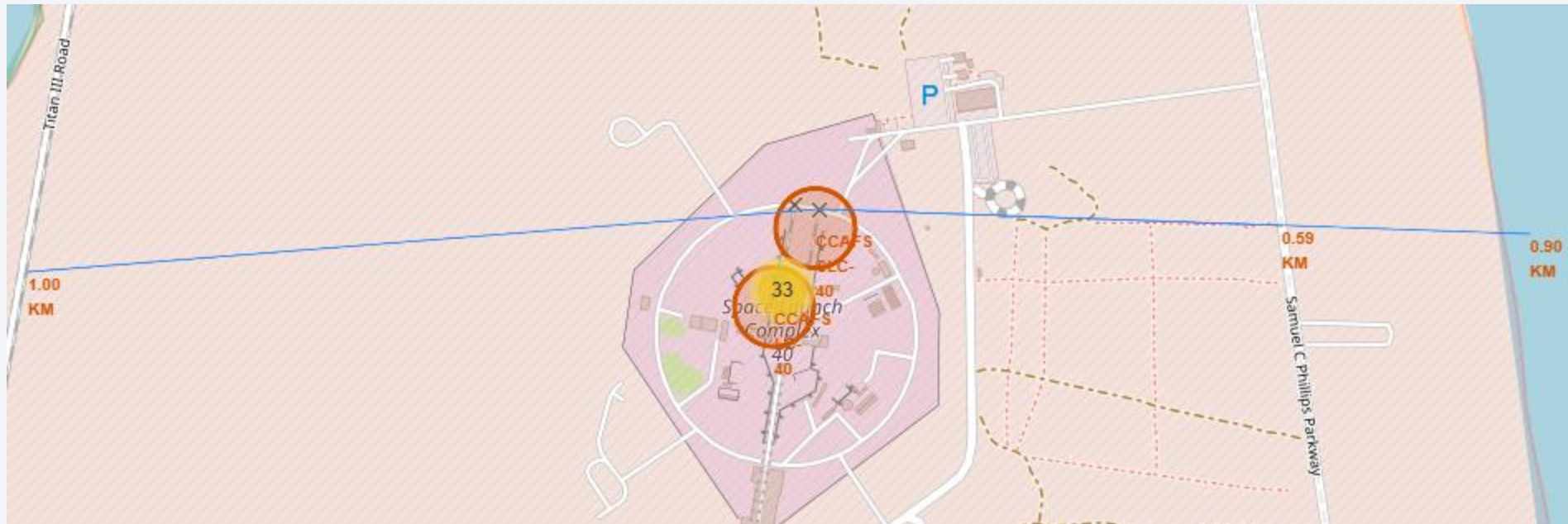
- East Coast Density: A high-density cluster of 46 points is visible near Florida.
- West Coast Density: A smaller cluster of 10 points is located near California.

Folium Map: Success rate for each launch outcome

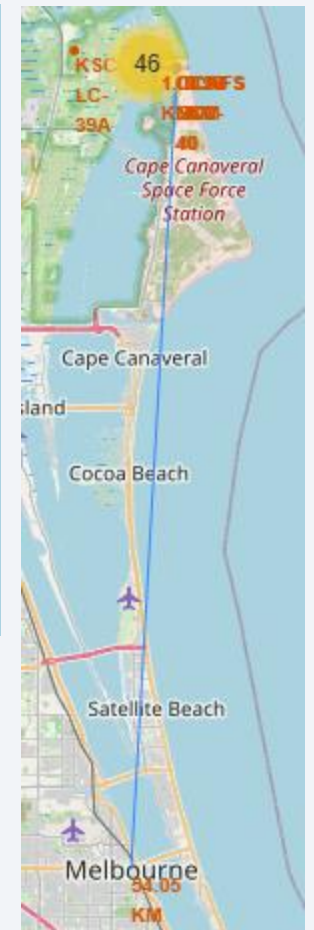
- From the color-labeled markers in marker clusters, KSC LC-39A has the higher success rate.
- **Green** Marker = **Successful** Return
- **Red** Marker = **Failed** Return

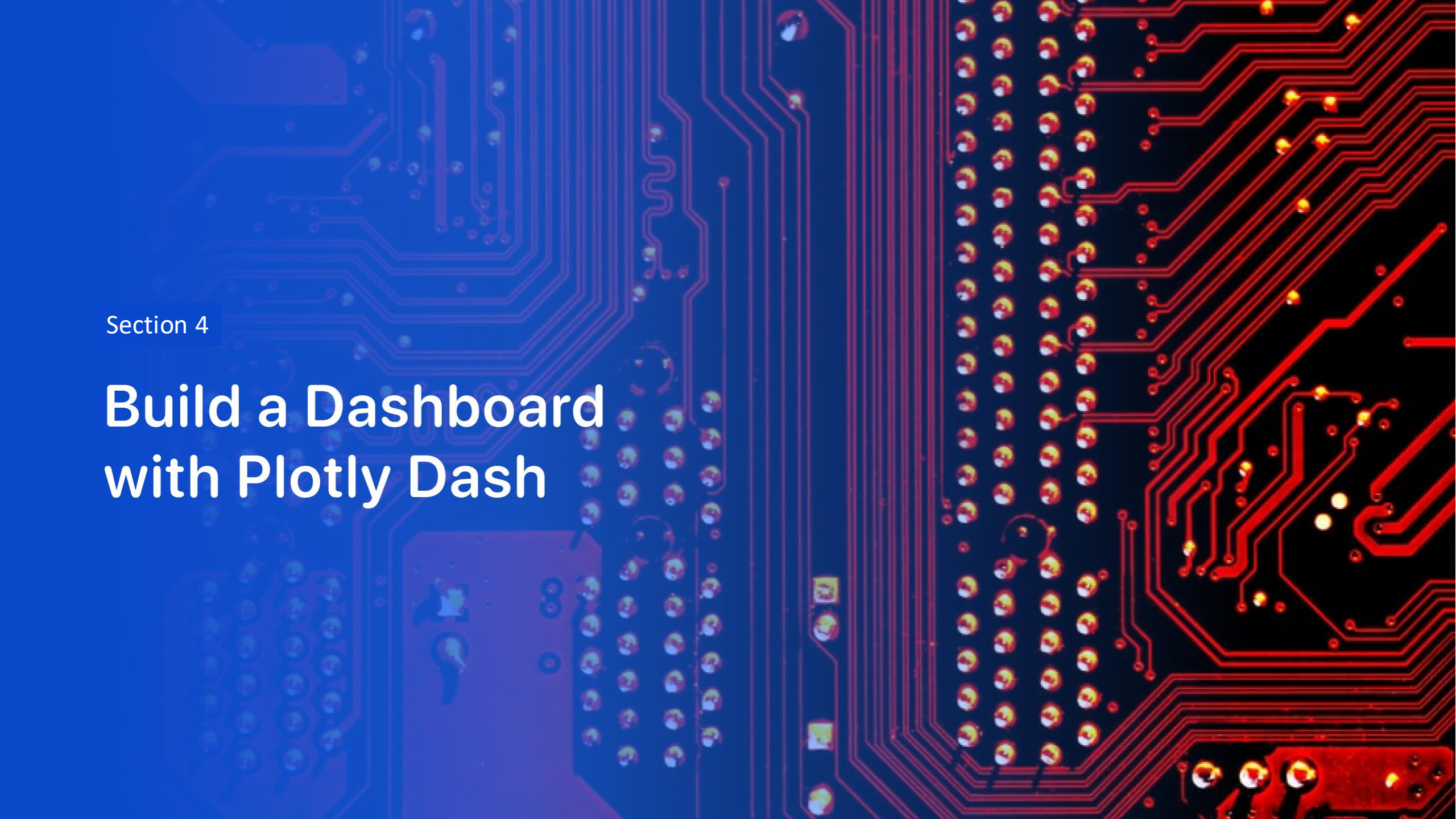


Folium Map: Closest Proximities to CCAFS SLC-40



Coastline Distance $\approx$ 0.90 KM
Railway Distance $\approx$ 1.00 KM
Highway Distance $\approx$ 0.59 KM
Melbourne Distance $\approx$ 54.05 KM

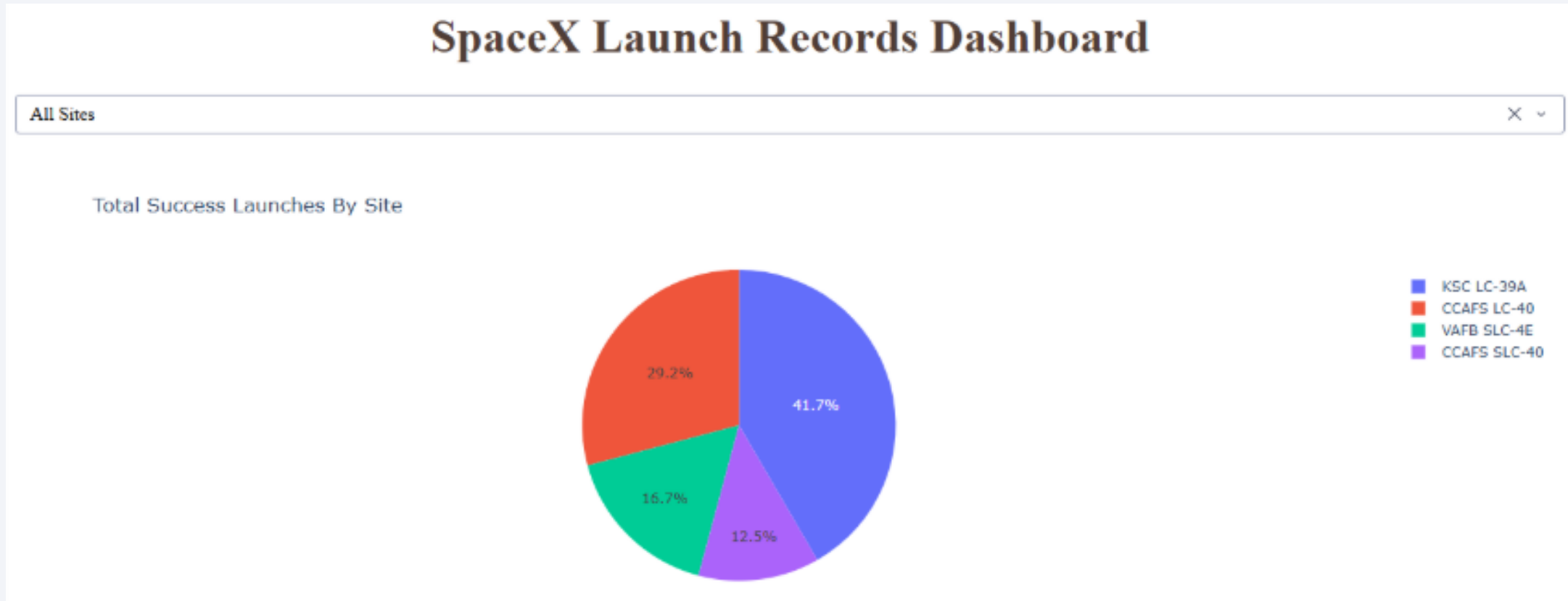




Section 4

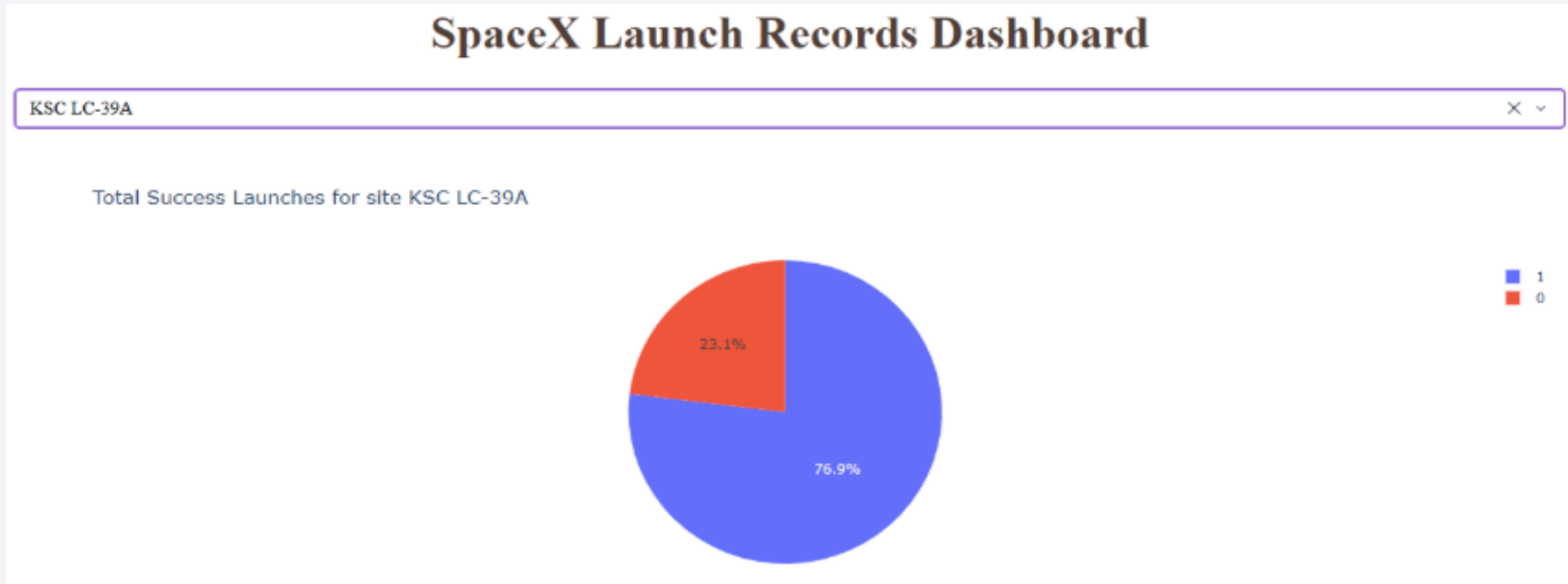
Build a Dashboard with Plotly Dash

Dashboard SpaceX Launch Records



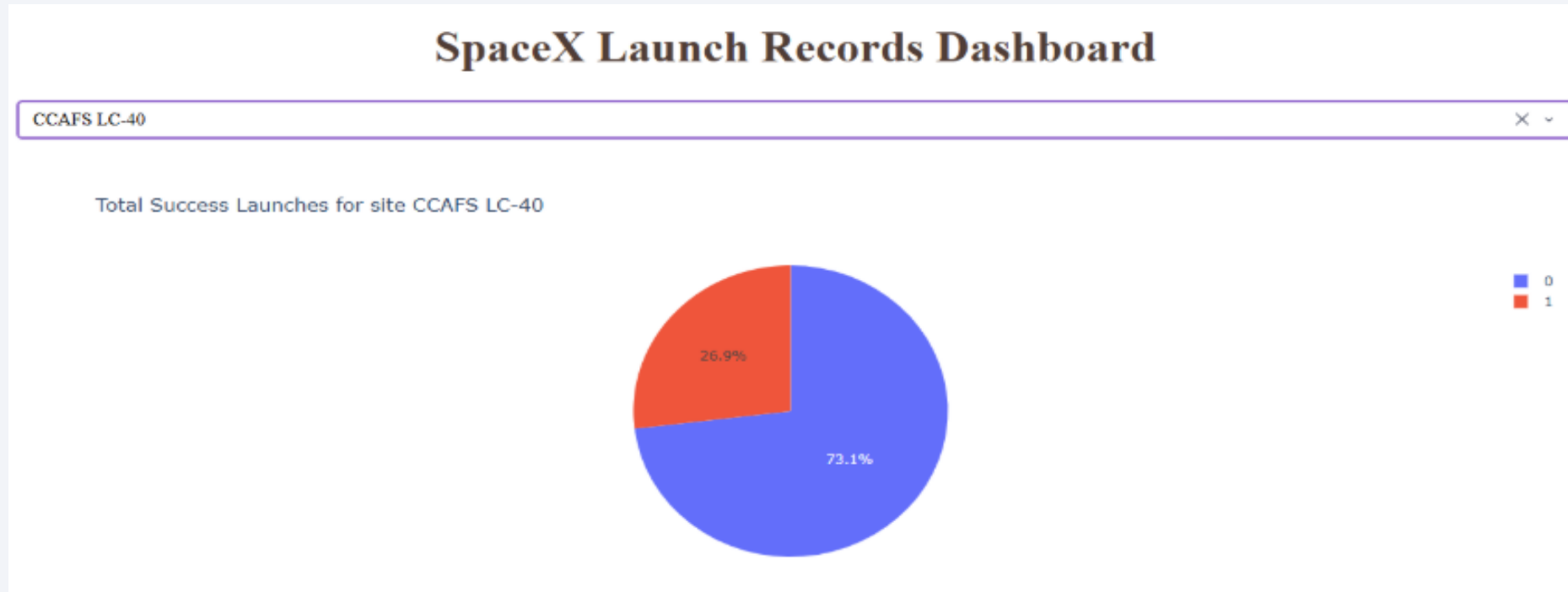
- This pie chart represents the total success launches for all sites, where KSC LC-39A has the largest successful launches with 41.7%.

Dashboard Total Success Launches for KSC LC-39A



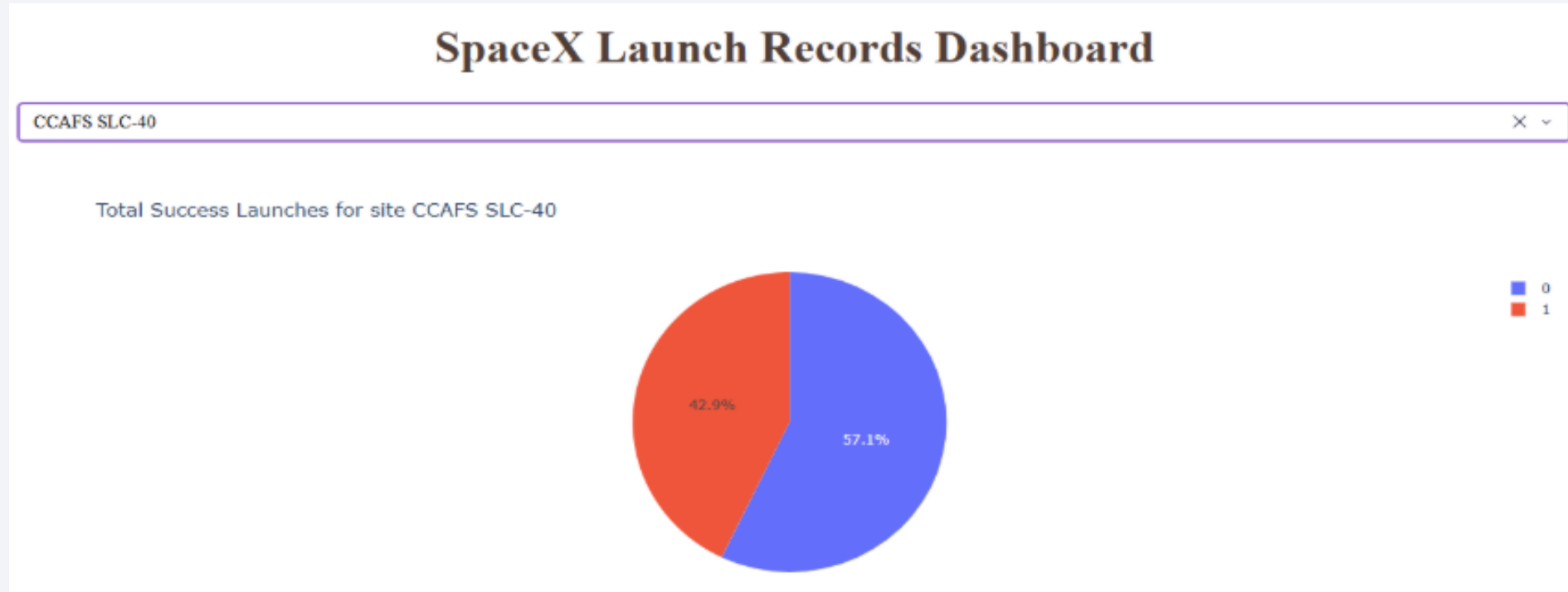
- This interactive pie chart represents the total success launches for site KSC LC-39A.
- KSC LC-39A also has the highest launch success rate of all launch sites, with a success rate of 76.9%.

Dashboard Total Success Launches for CCAFS LC-40



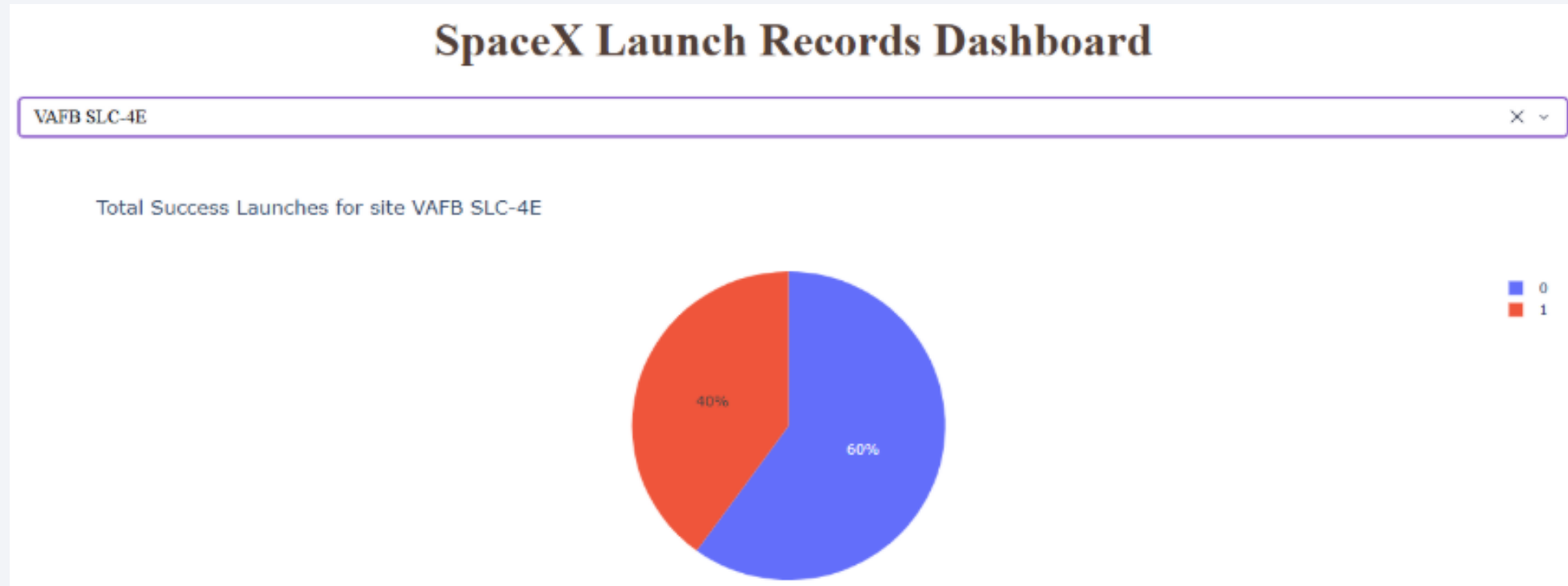
- This interactive pie chart represents the total success launches for site CCAFS LC-40.

Dashboard Total Success Launches for CCAFS SLC-40



- This interactive pie chart represents the total success launches for site CCAFS SLC-40.

Dashboard Total Success Launches for VAFB SLC-4E



- This interactive pie chart represents the total success launches for site VAFB SLC-4E.

Dashboard Payload vs Launch Outcome



- This interactive scatter plot shows the correlation between Payload and Success for all sites.
- The booster version that has the largest success rate is FT.

Dashboard Payload Range With Largest Success Rate



- The payload range that has the largest success rate is between 2000 kg to 4000 kg.

Dashboard Payload vs Launch Outcome For CCAFS LC-40



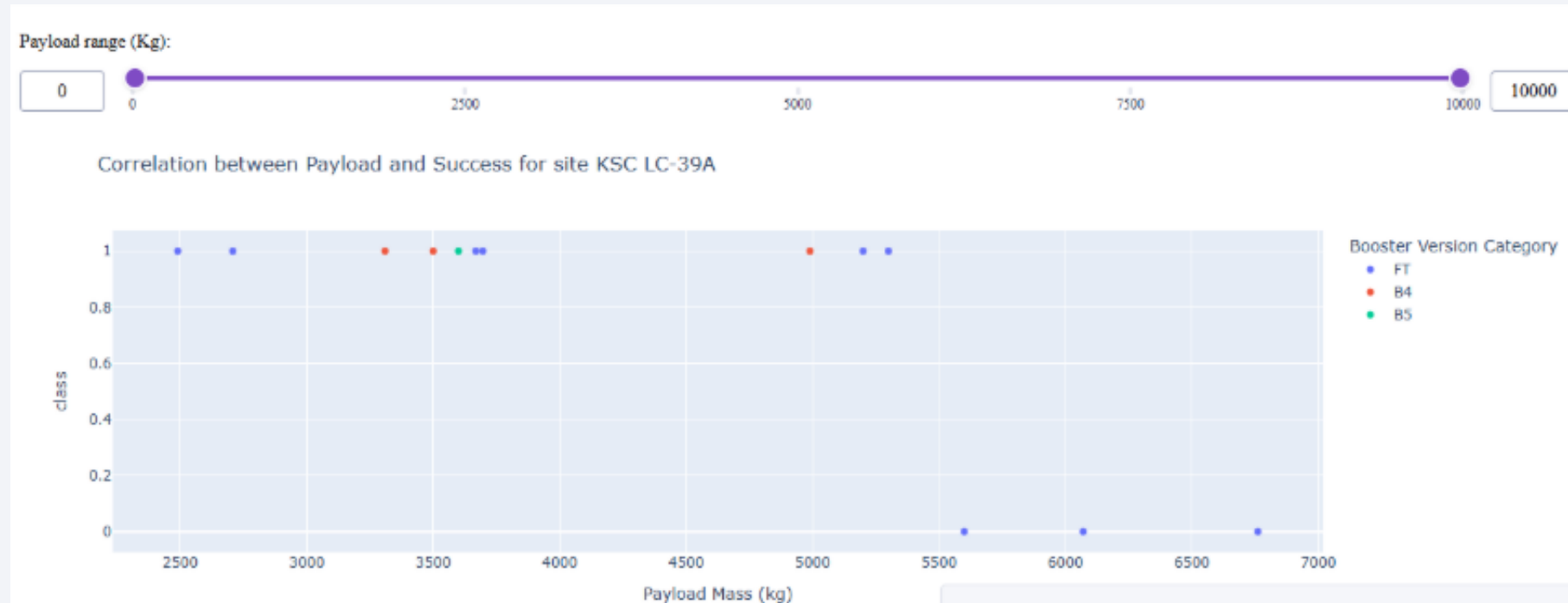
- This interactive scatter plot shows the correlation between Payload and Success for site CCAFS LC-40.

Dashboard Payload vs Launch Outcome For CCAFS SLC-40



- This interactive scatter plot shows the correlation between Payload and Success for site CCAFS SLC-40.

Dashboard Payload vs Launch Outcome For KSC LC-39A



- This interactive scatter plot shows the correlation between Payload and Success for site KSC LC-39A.

Dashboard Payload vs Launch Outcome For VAFB SLC-4E



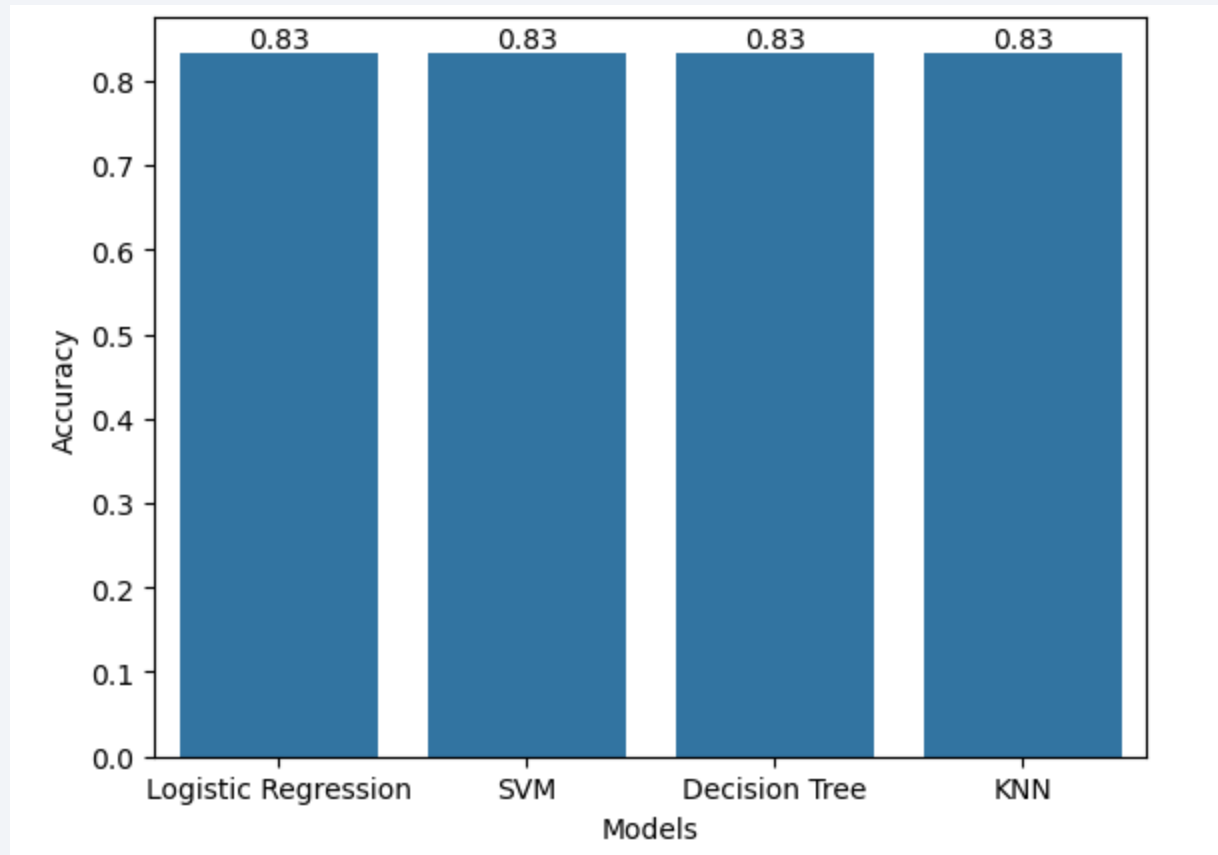
- This interactive scatter plot shows the correlation between Payload and Success for site VAFB SLC-4E.



Section 5

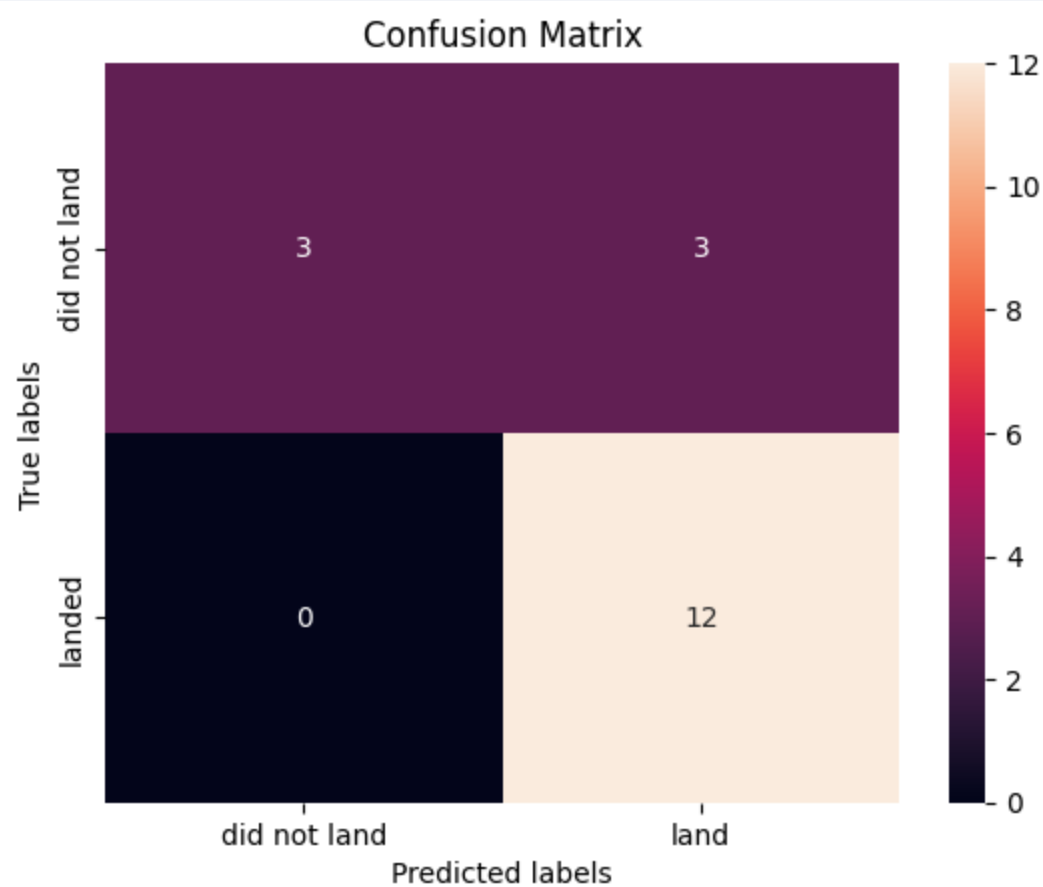
Predictive Analysis (Classification)

Classification Accuracy



- The four models (Logistic Regression, SVM, Decision Tree and KNN) has the same classification accuracy, so the accuracy on the test data was 0.83 in all models.

Confusion Matrix



- Logistic Regression, SVM, Decision Tree and KNN have the same confusion matrix. All the models can distinguish between the different classes, but there are three false positives that incorrectly predict a successful launch when it might fail.

True positive – 12 (True label is landed, Predicted label is also landed)

False positive – 3 (True label is not landed, Predicted label is landed)

True negative – 3 (True label is not landed, Predicted label is also not landed)

False negative – 0 (True label is landed, Predicted label is not landed)

Conclusions

- All launch sites with a higher volume of flights correlates with a success rate and orbits ES-L1, GEO, HEO and SSO had the most success rate.
- The launch sites are located in two US states: California and Florida, they are very close to the coast because it allows for safer rocket debris disposal and keep distance from cities because it minimizes risks to populated areas.
- KSC LC-39A has the highest launch success rate of all launch sites, with a success rate of 76.9%. This means that it achieved 10 successful launches out of 13 attempts (with only 3 failures).
- The payload range that has the largest success rate is between 2000 kg to 4000 kg. This means that the success rate for low weighted payloads is higher than the heavy weighted payloads.
- All the models has the same classification accuracy, whether you use Logistic Regression, SVM, Decision Tree or KNN, the result will be the same and while evaluating models, the main problem is the 3 false positives because they incorrectly predict a successful launch when it might fail. Future research should focus on adding more features because it would provide deeper patterns to the models or even changing the test data samples because it helps to verify if the results are consistent across different data splits and also evaluating the F1-Score, Precision and Recall because these metrics would offer a better understanding of the model's performance beyond accuracy.

Appendix

- SpaceX API URL ["Click Here"](#)
- SpaceX Wikipedia URL ["Click Here"](#)
- SpaceX data used in ML training ["Click Here"](#)

Thank you!

